

DATA SCIENCE PROJECT

Customer Segmentation for Targeted Retention and Value Growth

A robust RFM-based analysis of e-commerce purchase behaviour

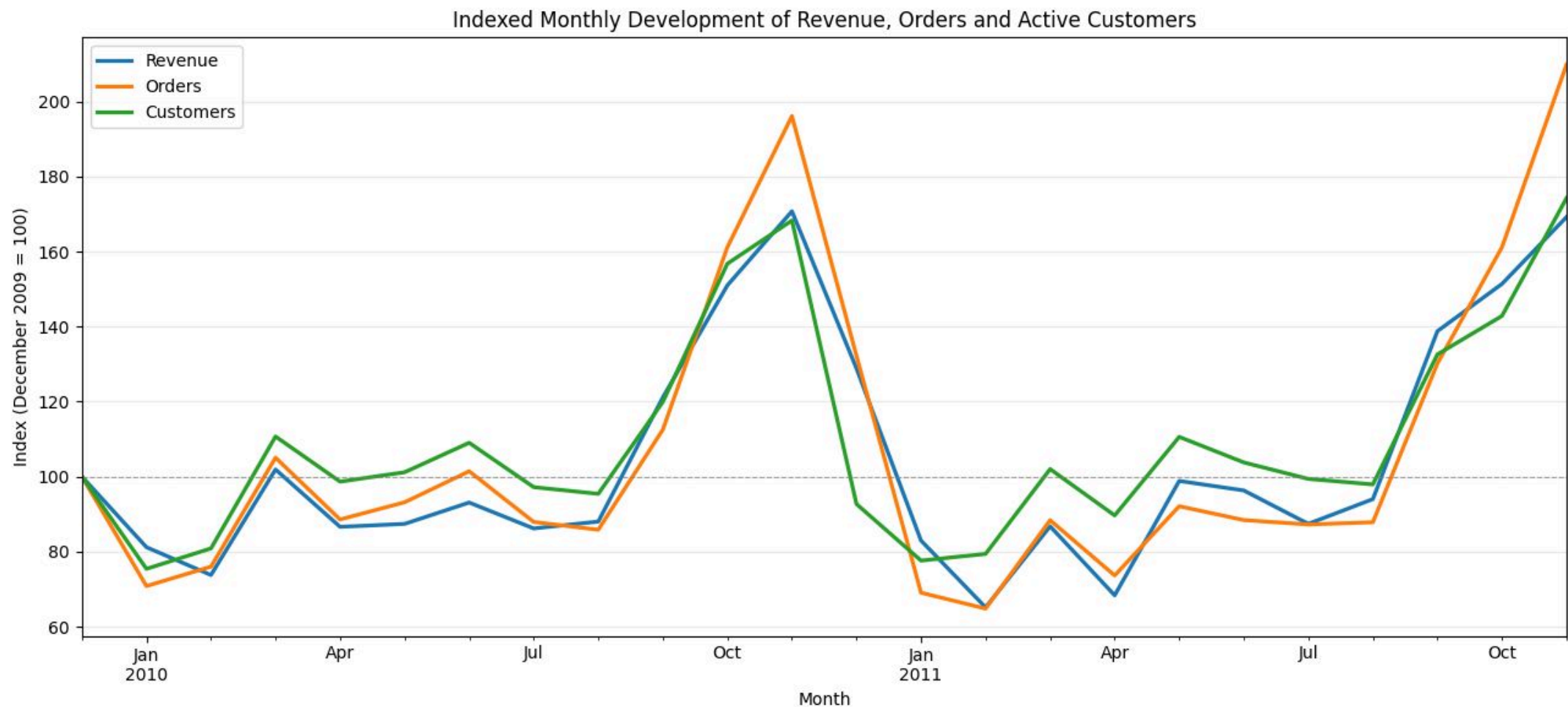
 **Presentation:** Heike Thöricht

 **Bremen**, 2 September 2026

Data source: UCI Online Retail II dataset (2009–2011); customer-level analysis based on 5,879 identifiable customers.

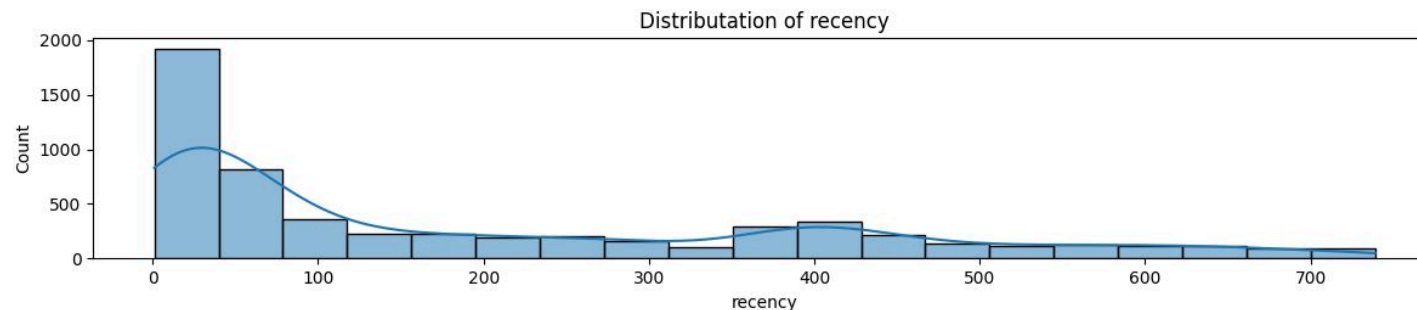
Customer value and engagement vary strongly

Most customers purchase infrequently and spend relatively little, while a small group generates disproportionately high order frequency and spend.



Why customer segmentation is needed

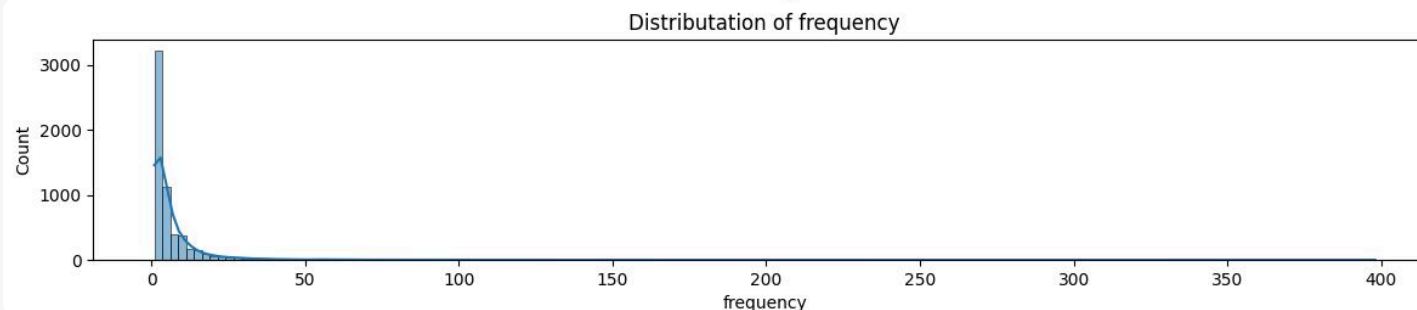
Distribution of customer behaviour across three key RFM dimensions — revealing how recently, how often, and how much each customer has purchased. Log transformation and p99 capping reduce the influence of extreme values before clustering.



Recency

Days since last purchase

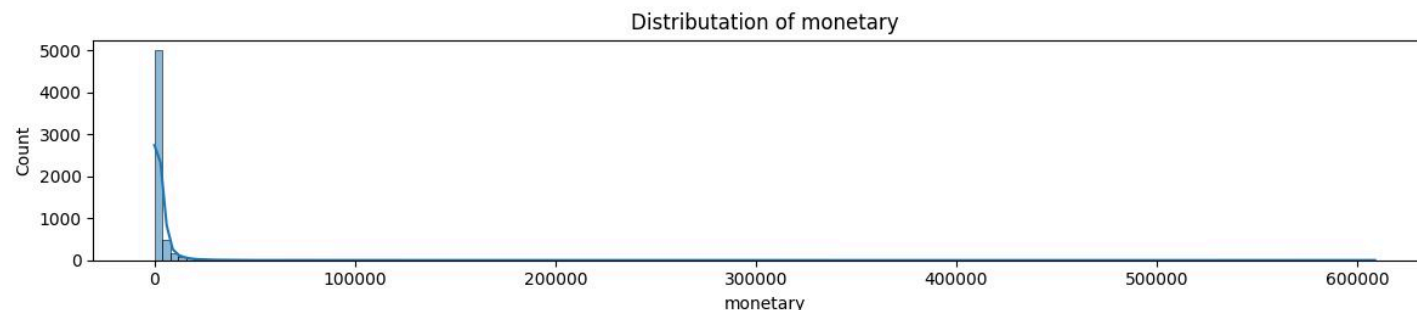
Right-skewed



Frequency

Number of purchases

Highly right-skewed



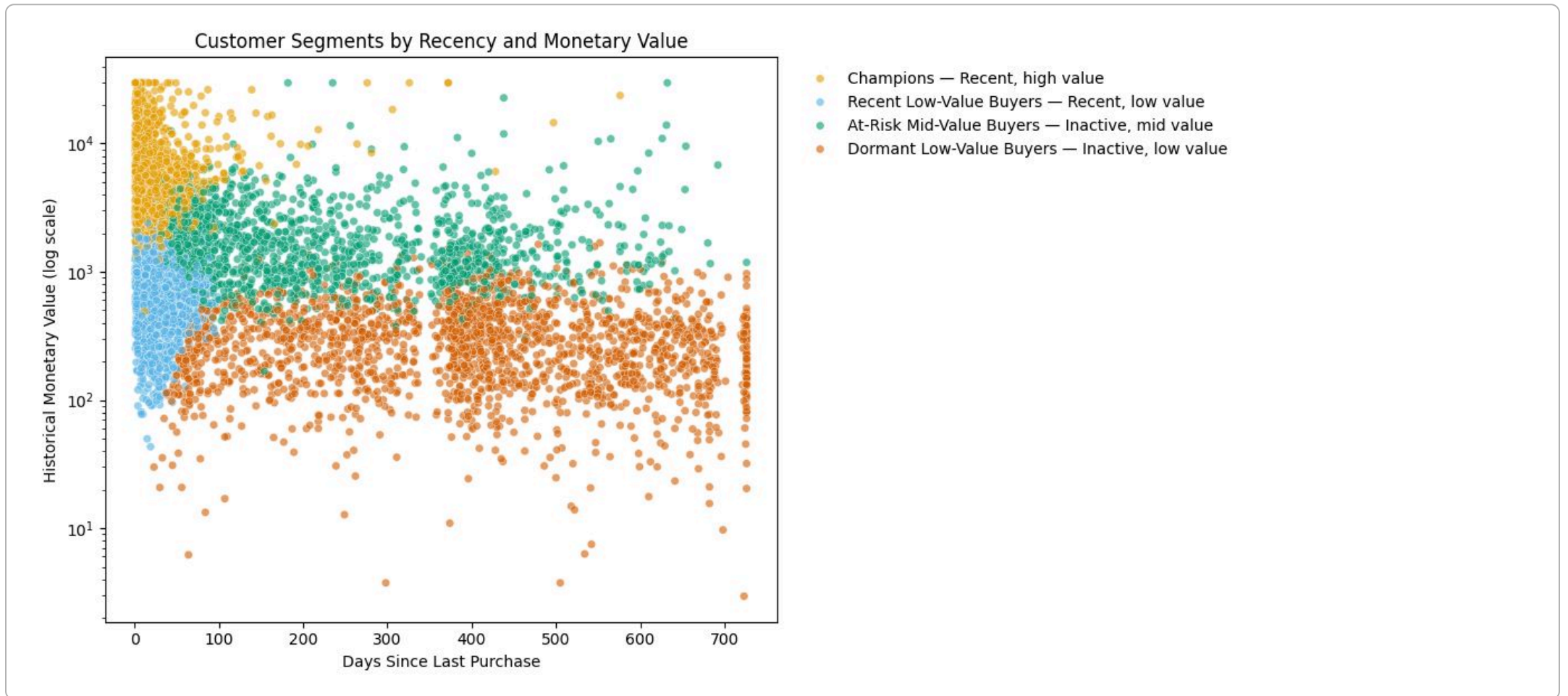
Monetary

Historical customer spend

Highly right-skewed

The observed heterogeneity indicates that customers should not be treated as one homogeneous group. We therefore use RFM-based clustering to identify actionable customer segments.

Four distinct customer segments emerge from RFM clustering



Segments are balanced, stable and actionable

The four segments differ clearly in purchase activity, repeat behaviour and customer value.

Customer segment	Customers	Share	Recency	Frequency	Monetary
Champions	1,219	20.8%	17 days	13 orders	5,067
Recent Low-Value Buyers	1,238	21.1%	24 days	3 orders	730
At-Risk Mid-Value Customers	1,457	24.8%	185 days	4 orders	1,496
Dormant Low-Value Customers	1,964	33.4%	405 days	1 order	282

Median values; n = 5,879 identifiable customers. Robust RFM clustering: silhouette score = 0.369; mean ARI = 0.980.

Segment-specific actions prioritise retention and customer value

The segmentation enables differentiated CRM actions instead of one-size-fits-all communication.

Champions

Goal: Retain and grow value

Action: VIP programme, cross-sell, personalised recommendations

Priority: HIGH

Recent Low-Value Buyers

Goal: Encourage repeat purchase

Action: Post-purchase journey, bundle or second-purchase incentive

Priority: MEDIUM-HIGH

At-Risk Mid-Value Customers

Goal: Win back historical value

Action: Personalised reactivation campaign

Priority: HIGH

Dormant Low-Value Customers

Goal: Limit cost of reactivation

Action: Low-cost automated outreach or suppression

Priority: LOW

- ❏ Priority focus: Retain Champions and reactivate At-risk customers; use scalable journeys to develop Recent low-value buyers and manage Dormant customers cost-effectively.

A robust RFM model provides the clearest customer segmentation

The four-cluster Robust RFM solution was selected for business use because it combined strong stability (mean ARI = 0.980; SD = 0.006), balanced segment sizes, and actionable customer profiles — with an acceptable silhouette score of 0.369.

Model	Best silhouette score	Assessment
Robust RFM	0.369	Best overall separation; selected core approach
Robust RFM + NLP	0.383	No meaningful improvement over RFM only model
Robust RFM + Behaviour	0.209	Behaviour features reduced cluster separation

Why NLP was not pursued:

- NLP features did not improve cluster separation relative to added complexity
- RFM-only model is simpler, more stable, and easier to operationalise
- Text-based insights are retained for segment profiling, not clustering

Customer segments show distinct seasonal purchase patterns

Temporal features were used for post-clustering profiling and campaign activation rather than for defining the core segments.

Segment	Autumn	Spring	Summer	Winter
At-Risk Mid-Value Customers	40.6%	26.4%	19.1%	13.9%
Champions	59.8%	13.9%	12.8%	13.5%
Dormant Low-Value Customers	36.6%	26.5%	18.6%	18.3%
Recent Low-Value Buyers	85.9%	4.8%	4.2%	5.2%

Recent buyers are overwhelmingly autumn-driven. Champions concentrate in autumn. At-risk and Dormant customers are more evenly distributed — enabling year-round reactivation campaigns.

Weekday peaks enable precise campaign timing

Segment	Mo	Di	Mi	Do	Fr	Sa	So
At-Risk Mid-Value Customers	24.2%	22.4%	17.8%	18.8%	7.6%	0.0%	9.1%
Champions	18.5%	20.3%	19.8%	22.9%	9.2%	0.0%	9.4%
Dormant Low-Value Customers	18.9%	19.5%	19.3%	18.4%	11.6%	0.1%	12.3%
Recent Low-Value Buyers	23.9%	22.6%	17.5%	17.0%	9.0%	0.0%	10.0%

Champions peak on Thursday. Recent and At-risk buyers concentrate on Monday/Tuesday. Saturday is negligible across all segments.

ADD ON

Additional insights for sales and marketing

Insight	Evidence	Implication
Midday is the dominant preferred period	49.4–71.6% across segments	Test campaign delivery around 10:00–14:00
Weekdays dominate	Saturday share is 0.0–0.1%; Sunday 9.1–12.3%	Prioritise weekday communication
Beginning of month is most common	40.1–48.1% across segments	Concentrate selected promotions early in the month
Autumn is strongest	36.6–85.9% across segments	Prioritise seasonal campaigns in autumn

Key takeaways and recommendations

→ Four actionable customer segments

Champions, At-Risk Mid-Value, Dormant Low-Value, and Recent Low-Value customers — each with a distinct RFM profile and a clear activation hypothesis.

→ Segment-specific strategies support retention and growth

Tailor messaging, offers, and channels to each segment's profile, then validate impact through controlled campaign experiments.

→ Temporal profiles inform campaign timing

Use observed purchase-time patterns — including midday, weekdays, early-month, and autumn concentration — to design and test more targeted campaign schedules.

→ Activate segments across relevant touchpoints

Use segment profiles across email, CRM, and paid media where customer identifiers and consent allow, while validating channel-specific performance.

From segmentation to testable activation

Use the four RFM segments as the foundation for differentiated CRM journeys. Combine them with temporal profiles to formulate campaign-timing hypotheses, and validate retention, reactivation, and growth effects through controlled experiments.